Homework 4

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Problem 1

Problem 1.1. *The metric for S^3 is*

$$\mathrm{d}s^2 = \mathrm{d}\theta_1^2 + \sin^2\theta_1\left(\mathrm{d}\theta_2^2 + \sin^2\theta_2 \mathrm{d}\theta_3^2\right).$$

Compute R.

Solution. The metric is $g_{\mu\nu} = \mathrm{diag}\left(1, \sin^2\theta_1, \sin^2\theta_1\sin^2\theta_2\right)$. Thus $g^{\mu\nu} = \mathrm{diag}\left(1, \csc^2\theta_1, \csc^2\theta_1\csc^2\theta_2\right)$. The Ricci scalar is given by $R = g^{\mu\nu}R_{\mu\nu}$. The Ricci tensor is given by

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu} = \partial_\lambda\Gamma^\lambda_{\nu\mu} - \partial_\nu\Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{\nu\mu} - \Gamma^\lambda_{\nu\alpha}\Gamma^\alpha_{\lambda\mu}.$$

It might be helpful to compute the Christoffel symbol. We do this with the action principal. We have

$$\mathcal{L} = g_{\mu\nu}\frac{\mathrm{d}\theta_\mu}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_\nu}{\mathrm{d}\tau} = g_{\mu\mu}\left(\frac{\mathrm{d}\theta_\mu}{\mathrm{d}\tau}\right)^2 = \dot\theta_1^2 + \sin^2\theta_1\dot\theta_2^2 + \sin^2\theta_1\sin^2\theta_2\dot\theta_3^2.$$

This satisfies the Euler-Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\mathcal{L}}{\partial\dot\theta_\mu} - \frac{\partial\mathcal{L}}{\partial\theta_\mu} = 0.$$

Note that I use the symbol τ for proper time even though there is no time because I didn't realize I should probably use σ for proper distance like last homework and have written to much to change it now. We compute the partials:

$$\frac{\partial\mathcal{L}}{\partial\theta_1} = 2\sin\theta_1\cos\theta_1\dot\theta_2^2 + 2\sin\theta_1\cos\theta_1\sin^2\theta_2\dot\theta_3^2$$

$$\frac{\partial\mathcal{L}}{\partial\theta_2} = 2\sin^2\theta_1\sin\theta_2\cos\theta_2\dot\theta_3^2$$

$$\frac{\partial\mathcal{L}}{\partial\theta_3} = 0$$

$$\frac{\partial\mathcal{L}}{\partial\dot\theta_1} = 2\dot\theta_1$$

$$\frac{\partial\mathcal{L}}{\partial\dot\theta_2} = 2\sin^2\theta_1\dot\theta_2$$

$$\frac{\partial\mathcal{L}}{\partial\dot\theta_3} = 2\sin^2\theta_1\sin^2\theta_2\dot\theta_3.$$

We now compute the relevant τ-derivatives:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\mathcal{L}}{\partial\dot\theta_1} = 2\ddot\theta_1$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} = 4\sin\theta_1\cos\theta_1\dot{\theta}_1\dot{\theta}_2 + 2\sin^2\theta_1\ddot{\theta}_2$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_3} &= 4\sin\theta_1\cos\theta_1\sin^2\theta_2\dot{\theta}_1\dot{\theta}_3 + 2\sin^2\theta_1\left(2\sin\theta_2\cos\theta_2\dot{\theta}_2\dot{\theta}_3 + \sin^2\theta_2\ddot{\theta}_3\right)\\ &= 4\sin\theta_1\cos\theta_1\sin^2\theta_2\dot{\theta}_1\dot{\theta}_3 + 4\sin^2\theta_1\sin\theta_2\cos\theta_2\dot{\theta}_2\dot{\theta}_3 + 2\sin^2\theta_1\sin^2\theta_2\ddot{\theta}_3.\end{aligned}$$

This yields the Euler-Lagrange equations

$$2\ddot{\theta}_1 - 2\sin\theta_1\cos\theta_1\dot{\theta}_2^2 - 2\sin\theta_1\cos\theta_1\sin^2\theta_2\dot{\theta}_3^2 = 0$$

$$4\sin\theta_1\cos\theta_1\dot{\theta}_1\dot{\theta}_2 + 2\sin^2\theta_1\ddot{\theta}_2 - 2\sin^2\theta_1\sin\theta_2\cos\theta_2\dot{\theta}_3^2 = 0$$

$$4\sin\theta_1\cos\theta_1\sin^2\theta_2\dot{\theta}_1\dot{\theta}_3 + 4\sin^2\theta_1\sin\theta_2\cos\theta_2\dot{\theta}_2\dot{\theta}_3 + 2\sin^2\theta_1\sin^2\theta_2\ddot{\theta}_3 = 0.$$

We must rearrange each of these equations to recover the geodesic equation

$$\frac{\mathrm{d}^2\theta_\mu}{\mathrm{d}\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}\theta_\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_\beta}{\mathrm{d}\tau},$$

from which we can then read off the Christoffel symbols.

Every single term in every single Euler-Lagrange equation has a factor of 2, so we divide out by that in each equation. We start with $\mu = 1$, which now reads

$$\frac{\mathrm{d}^2\theta_1}{\mathrm{d}\tau^2} + \left(-\sin\theta_1\cos\theta_1\left(\frac{\mathrm{d}\theta_2}{\mathrm{d}\tau}\right)^2 - \sin\theta_1\cos\theta_1\sin^2\theta_2\left(\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau}\right)^2\right).$$

From this we see that

$$\Gamma^1_{\alpha\beta} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\sin\theta_1\cos\theta_1 & 0 \\ 0 & 0 & -\sin\theta_1\cos\theta_1\sin^2\theta_2 \end{pmatrix}.$$

For the second equation, divide out by $\sin^2\theta_1$ to obtain

$$\frac{\mathrm{d}^2\theta_2}{\mathrm{d}\tau^2} + \left(\frac{2\cos\theta_1}{\sin\theta_1}\frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_2}{\mathrm{d}\tau} - \sin\theta_2\cos\theta_2\left(\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau}\right)^2\right) = 0.$$

The factor of 2 in the cotangent term comes from the fact that

$$\frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_2}{\mathrm{d}\tau} = \frac{\mathrm{d}\theta_2}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_1}{\mathrm{d}\tau},$$

meaning that this term really represents

$$\cot\theta_1\frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_2}{\mathrm{d}\tau} + \cot\theta_1\frac{\mathrm{d}\theta_2}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}.$$

We can see this also by symmetry of the lower indices in $\Gamma^\mu_{\alpha\beta}$. This yields

$$\Gamma^2_{\alpha\beta} = \begin{pmatrix} 0 & \cot\theta_1 & 0 \\ \cot\theta_1 & 0 & 0 \\ 0 & 0 & -\sin\theta_2\cos\theta_2 \end{pmatrix}.$$

For the final equation, we have

$$\sin^2\theta_1 \sin^2\theta_2 \frac{\mathrm{d}^2\theta_3}{\mathrm{d}\tau^2} + \left(2\sin\theta_1\cos\theta_1\sin^2\theta_2 \frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau} + 2\sin^2\theta_1\sin\theta_2\cos\theta_2 \frac{\mathrm{d}\theta_2}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau}\right) = 0.$$

Dividing out by $\sin^2\theta_1 \sin^2\theta_2$ yields

$$\frac{\mathrm{d}^2\theta_3}{\mathrm{d}\tau^2} + \left(2\cot\theta_1 \frac{\mathrm{d}\theta_1}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau} + 2\cot\theta_2 \frac{\mathrm{d}\theta_2}{\mathrm{d}\tau}\frac{\mathrm{d}\theta_3}{\mathrm{d}\tau}\right).$$

We then obtain by the same logic as before that

$$\Gamma^3_{\alpha\beta} = \begin{pmatrix} 0 & 0 & \cot\theta_1 \\ 0 & 0 & \cot\theta_2 \\ \cot\theta_1 & \cot\theta_2 & 0 \end{pmatrix}.$$

Now return to our computation of $R_{\mu\nu}$. We must compute the four terms of the equation

$$R^\lambda{}_{\mu\lambda\nu} = \partial_\lambda \Gamma^\lambda_{\nu\mu} - \partial_\nu \Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{\nu\mu} - \Gamma^\lambda_{\nu\alpha}\Gamma^\alpha_{\lambda\mu}.$$

We can simplify this computation by noting that for our purposes of comptuing R, we need not compute the entire Ricci tensor. Since $g_{\mu\nu} = g_{\mu\mu}$ since the metri is diagonal, we have

$$R = g^{\mu\nu} R_{\mu\nu} = g^{\mu\mu} R_{\mu\mu}.$$

We thus only need to compute the diagonal components, so we need only compute

$$R_{\mu\mu} = R^\lambda{}_{\mu\lambda\mu} = \partial_\lambda \Gamma^\lambda_{\mu\mu} - \partial_\mu \Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{\mu\mu} - \Gamma^\lambda_{\mu\alpha}\Gamma^\alpha_{\lambda\mu}.$$

Start by taking $\mu = 1$. We compute each term independently. First the terms with partials:

$$\partial_\lambda \Gamma^\lambda_{11} = \frac{\partial}{\partial\theta_1} 0 + \frac{\partial}{\partial\theta_2}(-\sin\theta_1\cos\theta_1) + \frac{\partial}{\partial\theta_3}\left(-\sin\theta_1\cos\theta_1\sin^2\theta_2\right) = 0$$

$$\partial_1 \Gamma^\lambda_{\lambda 1} = \frac{\partial}{\partial\theta_1}\Gamma^1_{11} + \frac{\partial}{\partial\theta_1}\Gamma^2_{21} + \frac{\partial}{\partial\theta_1}\Gamma^3_{31} = 0 + 2\frac{\partial}{\partial\theta_1}\cot\theta_1 = -2\csc^2\theta_1.$$

This next term follows since $\Gamma^\alpha_{11} = 0$ for all α:

$$\Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{11} = 0.$$

Finally,

$$\begin{aligned} \Gamma^\lambda_{1\alpha}\Gamma^\alpha_{\lambda 1} &= \Gamma^1_{11}\Gamma^1_{11} + \Gamma^1_{12}\Gamma^2_{11} + \Gamma^1_{13}\Gamma^3_{11} + \Gamma^2_{11}\Gamma^1_{21} + \Gamma^2_{12}\Gamma^2_{21} + \Gamma^2_{13}\Gamma^3_{21} + \Gamma^3_{11}\Gamma^1_{31} + \Gamma^3_{12}\Gamma^2_{31} + \Gamma^3_{13}\Gamma^3_{31} \\ &= 0 + 0 + 0 + 0 + \cot^2\theta_1 + 0 + 0 + 0 + \cot^2\theta_1 = 2\cot^2\theta_1. \end{aligned}$$

Therefore,

$$R_{11} = 0 + 2\csc^2\theta_1 + 0 - 2\cot^2\theta_1 = 2\left(\csc^2\theta_1 - \cot^2\theta_1\right) = 2\left(\frac{1}{\sin^2\theta_1} - \frac{\cos^2\theta_1}{\sin^2\theta_1}\right) = 2\frac{\sin^2\theta_1}{\sin^2\theta_1} = 2.$$

Next take $\mu = 2$. We once again start with the partials:

$$\partial_\lambda \Gamma^\lambda_{22} = \frac{\partial}{\partial\theta_1}\Gamma^1_{22} + \frac{\partial}{\partial\theta_2}\Gamma^2_{22} + \frac{\partial}{\partial\theta_3}\Gamma^3_{22} = \frac{\partial}{\partial\theta_1}(-\sin\theta_1\cos\theta_1) = \sin^2\theta_1 - \cos^2\theta_1$$

$$\partial_2\Gamma^\lambda_{\lambda 2} = \frac{\partial}{\partial\theta_2}\Gamma^1_{12} + \frac{\partial}{\partial\theta_2}\Gamma^2_{22} + \frac{\partial}{\partial\theta_2}\Gamma^3_{32} = \frac{\partial}{\partial\theta_2}\cot\theta_2 = -\csc^2\theta_2.$$

Since Γ^α_{22} is nonzero only when $\alpha = 1$, the third term is once again simplified to

$$\Gamma^\lambda_{\lambda 1}\Gamma^1_{22} = -\sin\theta_1\cos\theta_1\left(\Gamma^1_{11} + \Gamma^2_{21} + \Gamma^3_{31}\right) = -\sin\theta_1\cos\theta_1\left(\cot\theta_1 + \cot\theta_1\right) = -2\sin\theta_1\cos\theta_1\frac{\cos\theta_1}{\sin\theta_1}$$

$$= -2\cos^2\theta_1.$$

$$\begin{aligned}\Gamma^\lambda_{2\alpha}\Gamma^\alpha_{\lambda 2} &= \Gamma^1_{21}\Gamma^1_{12} + \Gamma^1_{22}\Gamma^2_{12} + \Gamma^1_{23}\Gamma^3_{12} + \Gamma^2_{21}\Gamma^1_{22} + \Gamma^2_{22}\Gamma^2_{22} + \Gamma^2_{23}\Gamma^3_{22} + \Gamma^3_{21}\Gamma^1_{32} + \Gamma^3_{22}\Gamma^2_{32} + \Gamma^3_{23}\Gamma^3_{32}\\ &= 0 + -\sin\theta_1\cos\theta_1\cot\theta_1 + 0 - \cot\theta_1\sin\theta_1\cos\theta_1 + 0 + 0 + 0 + 0 + \cot^2\theta_2\\ &= \cot^2\theta_2 - 2\sin\theta_1\cos\theta_1\cot\theta_1 = \cot^2\theta_2 - 2\cos^2\theta_1.\end{aligned}$$

We then have

$$R_{22} = \sin^2\theta_1 - \cos^2\theta_1 + \csc^2\theta_2 - 2\cos^2\theta_1 - \cot^2\theta_2 + 2\cos^2\theta_1 = \sin^2\theta_1 - \cos^2\theta_1 + \frac{1}{\sin^2\theta_2} - \frac{\cos^2\theta_2}{\sin^2\theta_2}$$

$$= 1 - \cos^2\theta_1 + \sin^2\theta_1 = 2\sin^2\theta_1.$$

Finally, take $\mu = 3$.

$$\partial_\lambda\Gamma^\lambda_{33} = \frac{\partial}{\partial\theta_1}\Gamma^1_{33} + \frac{\partial}{\partial\theta_2}\Gamma^2_{33} + \frac{\partial}{\partial\theta_3}\Gamma^3_{33} = -\frac{\partial}{\partial\theta_1}\sin\theta_1\cos\theta_1\sin^2\theta_2 - \frac{\partial}{\partial\theta_2}\sin\theta_2\cos\theta_2$$

$$= -\sin^2\theta_2\left(\cos^2\theta_1 - \sin^2\theta_1\right) - \left(\cos^2\theta_2 - \sin^2\theta_2\right).$$

Because the Christoffels have no θ_3-dependence,

$$\partial_3\Gamma^\lambda_{\lambda 3}\frac{\partial}{\partial\theta_3}\Gamma^1_{13} + \frac{\partial}{\partial\theta_3}\Gamma^2_{23} + \frac{\partial}{\partial\theta_3}\Gamma^3_{33} = 0.$$

For the next term, note that once again $\Gamma^\alpha_{33} \neq 0$ if and only if $\alpha = 1$ or 2, so it reduces to

$$\Gamma^\lambda_{\lambda 1}\Gamma^1_{33} = -\sin\theta_1\cos\theta_1\sin^2\theta_2\left(\Gamma^1_{11} + \Gamma^2_{21} + \Gamma^3_{31}\right) = -2\sin\theta_1\cos\theta_1\cot\theta_1\sin^2\theta_2$$

$$= -2\cos^2\theta_1\sin^2\theta_2$$

and

$$\Gamma^\lambda_{\lambda 2}\Gamma^2_{33} = -\sin\theta_2\cos\theta_2\left(\Gamma^1_{12} + \Gamma^2_{22} + \Gamma^3_{32}\right) = -\sin\theta_2\cos\theta_2\cot\theta_2 = -\cos^2\theta_2.$$

Finally,

$$\begin{aligned}\Gamma^\lambda_{3\alpha}\Gamma^\alpha_{\lambda 3} &= \Gamma^1_{31}\Gamma^1_{13} + \Gamma^1_{32}\Gamma^2_{13} + \Gamma^1_{33}\Gamma^3_{13} + \Gamma^2_{31}\Gamma^1_{23} + \Gamma^2_{32}\Gamma^2_{23} + \Gamma^2_{33}\Gamma^3_{23} + \Gamma^3_{31}\Gamma^1_{33} + \Gamma^3_{32}\Gamma^2_{33} + \Gamma^3_{33}\Gamma^3_{33}\\ &= 0 + 0 + \Gamma^1_{33}\Gamma^3_{13} + 0 + 0 + \Gamma^2_{33}\Gamma^3_{23} + \Gamma^3_{31}\Gamma^1_{33} + \Gamma^3_{32}\Gamma^2_{33} + 0\\ &= -\sin\theta_1\cos\theta_1\sin^2\theta_2\cot\theta_1 - \sin\theta_2\cos\theta_2\cot\theta_2 - \sin\theta_1\cos\theta_1\sin^2\theta_2\cot\theta_1 - \sin\theta_2\cos\theta_2\cot\theta_2\\ &= -2\left(\sin\theta_1\cos\theta_1\sin^2\theta_2\cot\theta_1 + \sin\theta_2\cos\theta_2\cot\theta_2\right) = -2\left(\cos^2\theta_1\sin^2\theta_2 + \cos^2\theta_2\right).\end{aligned}$$

We now have

$$R_{33} = -\sin^2\theta_2\left(\cos^2\theta_1 - \sin^2\theta_1\right) - \left(\cos^2\theta_2 - \sin^2\theta_2\right) - 2\cos^2\theta_1\sin^2\theta_2 - \cos^2\theta_2 + 2\left(\cos^2\theta_1\sin^2\theta_2 + \cos^2\theta_2\right)$$

$$= -\sin^2\theta_2\cos^2\theta_1 + \sin^2\theta_2\sin^2\theta_1 - \cos^2\theta_2 + \sin^2\theta_2 - 2\cos^2\theta_1\sin^2\theta_2 - \cos^2\theta_2 + 2\cos^2\theta_1\sin^2\theta_2 + 2\cos^2\theta_2$$

$$= -\cos^2\theta_1 \sin^2\theta_2 + \sin^2\theta_1 \sin^2\theta_2 + \sin^2\theta_2 = \sin^2\theta_2 \left(1 - \cos^2\theta_1\right) + \sin^2\theta_1 \sin^2\theta_2$$
$$= 2\sin^2\theta_1 \sin^2\theta_2.$$

We now have

$$R_{\mu\mu} = \left(2, 2\sin^2\theta_1, 2\sin^2\theta_1\sin^2\theta_2\right) = 2\left(1, \sin^2\theta_1, \sin^2\theta_1\sin^2\theta_2\right) = 2g_{\mu\mu}.$$

It follows that

$$R = g^{\mu\mu}R_{\mu\mu} = g^{\mu\mu}2g_{\mu\mu} = 2 + 2 + 2 = 6.$$

Therefore, $R = 6$.

Problem 1.2. *Now consider an expanding 3-spherical universe*

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\left(\mathrm{d}\theta_1^2 + \sin^2\theta_1\left(\mathrm{d}\theta_2^2 + \sin^2\theta_2\mathrm{d}\theta_3^2\right)\right).$$

Compute the Ricci scalar.

Solution. To simplify notation, we use the convention that roman letters as indices i, j, k run over spatial coordiantes $\theta_1, \theta_2, \theta_3$, and greek letters μ, ν, ρ run over spacetime coordinates $t, \theta_1, \theta_2, \theta_3$. Write h_{ij} for the metric used in problem 1.1. Then $g_{ij} = a^2(t)h_{ij}$, and

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = -\mathrm{d}t^2 + a^2(t)h_{ij}\mathrm{d}\theta_i\mathrm{d}\theta_j.$$

Since the metric is still diagonal, we can use the same simplification as last time and only compute $R_{\mu\mu}$. It would be nice if we could use the result of problem 1.1 to simplify the problem, so write

$$R_{\mu\mu} = R_{tt} + R_{ii}.$$

We will start by writing

$$R_{ii} = \partial_\lambda\Gamma^\lambda_{ii} - \partial_i\Gamma^\lambda_{\lambda i} + \Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{ii} - \Gamma^\lambda_{i\alpha}\Gamma^\alpha_{\lambda i}$$

in terms of the Ricci tensor from problem 1.1 $2h_{ii}$, which we will denote by K_{ii}. It will be useful to split this into a spatial component which ranges only over spatial coordinates, and then bundle the time-dependent terms into a different component. The relationship will then become more clear. We have

$$\begin{aligned} R_{ii} &= \partial_t\Gamma^t_{ii} - \partial_i\Gamma^t_{ti} + \Gamma^t_{t\alpha}\Gamma^\alpha_{ii} - \Gamma^t_{i\alpha}\Gamma^\alpha_{ti} + \partial_j\Gamma^j_{ii} - \partial_i\Gamma^j_{ji} + \Gamma^j_{j\alpha}\Gamma^\alpha_{ii} - \Gamma^j_{i\alpha}\Gamma^\alpha_{ji} \\ &= \partial_t\Gamma^t_{ii} - \partial_i\Gamma^t_{ti} + \Gamma^t_{t\alpha}\Gamma^\alpha_{ii} - \Gamma^t_{i\alpha}\Gamma^\alpha_{ti} + \partial_j\Gamma^j_{ii} - \partial_i\Gamma^j_{ji} + \Gamma^j_{jt}\Gamma^t_{ii} + \Gamma^j_{jk}\Gamma^k_{ii} - \Gamma^j_{it}\Gamma^t_{ji} - \Gamma^j_{ik}\Gamma^k_{ji}. \end{aligned}$$

Define the time component

$$T_{ii} = \partial_t\Gamma^t_{ii} - \partial_i\Gamma^t_{ti} + \Gamma^t_{t\alpha}\Gamma^\alpha_{ii} - \Gamma^t_{i\alpha}\Gamma^\alpha_{ti} + \Gamma^j_{jt}\Gamma^t_{ii} - \Gamma^j_{it}\Gamma^t_{ji}$$

and the space component

$$S_{ii} = \partial_j\Gamma^j_{ii} - \partial_i\Gamma^j_{ji} + \Gamma^j_{jk}\Gamma^k_{ii} - \Gamma^j_{ik}\Gamma^k_{ij}.$$

The form of the space component looks quite a bit like the Ricci tensor K_{ii} from problem 1.1. To find the exact relationship, we use the coordinate invariant definition of the Christoffel symbol

$$\Gamma^\rho_{\alpha\beta} = \frac{1}{2}g^{\sigma\rho}\left(\frac{\partial g_{\alpha\sigma}}{\partial x^\beta} + \frac{\partial g_{\beta\sigma}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma}\right).$$

Using $g_{ij} = a^2(t)h_{ij}$ and $g^{ij} = a^{-2}(t)h^{ij}$, and noting that $a^{-2}(t)$ has no spatial dependence and thus factors out of the partial derivatives,

$$\Gamma^i_{jk} = \frac{1}{2}g^{\ell i}\left(\frac{\partial g_{j\ell}}{\partial x^k} + \frac{\partial g_{k\ell}}{\partial x^j} - \frac{\partial g_{jk}}{\partial x^\ell}\right) = \frac{1}{2}\frac{h^{ij}}{a^2(t)}\left(a^2(t)\frac{\partial h_{j\ell}}{\partial x^k} + a^2(t)\frac{\partial h_{k\ell}}{\partial x^j} - a^2(t)\frac{\partial h_{jk}}{\partial x^\ell}\right)$$
$$= \frac{1}{2}h^{ij}\left(\frac{\partial h_{j\ell}}{\partial x^k} + \frac{\partial h_{k\ell}}{\partial x^j} - \frac{\partial h_{jk}}{\partial x^\ell}\right).$$

Note that this is exactly the same as the Christoffel symbol from the metric in problem 1.1, so we immediately see that $S_{ii} = K_{ii} = 2h_{ii}$. We now need only compute T_{ii} and R_{tt}.

We still need to compute some new Christoffel symbols, but since we only need to compute a few of them, we will not set up all of the Euler-Lagrange equations. It will be faster to use Euler-Lagrange for $\Gamma^t_{\mu\nu}$, and then to compute the remaining components directly from the definition. We have

$$\mathcal{L} = g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = -\dot{t}^2 + a^2(t)\dot{\theta}_1^2 + a^2(t)\sin^2\theta_1\dot{\theta}_2^2 + a^2(t)\sin^2\theta_1\sin^2\theta_2\dot{\theta}_3^2.$$

Then

$$\frac{\partial\mathcal{L}}{\partial\dot{t}} = -2\dot{t} \implies \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\mathcal{L}}{\partial\dot{t}} = -2\ddot{t},$$
$$\frac{\partial\mathcal{L}}{\partial t} = \frac{\partial}{\partial t}\left(-\dot{t}^2 + a^2(t)\left(h_{ii}\dot{\theta}_i^2\right)\right) = 2a\dot{a}h_{ii}\dot{\theta}_i^2.$$

Divding out by -2, the geodesic equation then takes the form

$$\frac{\mathrm{d}^2t}{\mathrm{d}\tau^2} + a\dot{a}h_{ii}\left(\frac{\mathrm{d}\theta_i}{\mathrm{d}\tau}\right)^2 = 0.$$

The Christoffel symbol is thus

$$\Gamma^t_{ij} = \Gamma^t_{ii} = a\dot{a}h_{ii}.$$

In particular,

$$\Gamma^t_{tt} = 0, \qquad \Gamma^t_{t\alpha} = \Gamma^t_{tt} = 0, \qquad \Gamma^t_{i\alpha} = \Gamma^t_{ii}, \qquad \Gamma^t_{ii} = a\dot{a}h_{ii}.$$

This will simplify our computation of the remaining components of the Ricci tensor. For instance, the time component

$$T_{ii} = \partial_t\Gamma^t_{ii} - \partial_i\Gamma^t_{ti} + \Gamma^t_{t\alpha}\Gamma^\alpha_{ii} - \Gamma^t_{i\alpha}\Gamma^\alpha_{ti} + \Gamma^j_{jt}\Gamma^t_{ii} - \Gamma^j_{it}\Gamma^t_{ji}$$

simplifies:

$$\partial_t\Gamma^t_{ii} = \partial_t a\dot{a}h_{ii} = \left(\dot{a}^2 + a\ddot{a}\right)h_{ii}$$
$$\partial_i\Gamma^t_{ti} = 0$$
$$\Gamma^t_{t\alpha}\Gamma^\alpha_{ii} = 0$$
$$\Gamma^t_{i\alpha}\Gamma^\alpha_{ti} = \Gamma^t_{ii}\Gamma^i_{ti} = a\dot{a}h_{ii}\Gamma^i_{ti}$$
$$\Gamma^j_{jt}\Gamma^t_{ii} = \Gamma^j_{jt}a\dot{a}h_{ii}$$
$$\Gamma^j_{it}\Gamma^t_{ji} = \Gamma^i_{it}\Gamma^t_{ii} = \Gamma^i_{it}a\dot{a}h_{ii}.$$

We then have

$$T_{ii} = \left(\dot{a}^2 + a\ddot{a}\right)h_{ii} - a\dot{a}h_{ii}\Gamma^i_{ti} + \Gamma^j_{jt}a\dot{a}h_{ii} - \Gamma^i_{it}a\dot{a}h_{ii}$$

$$= \left(\dot{a}^2 + a\ddot{a}\right) h_{ii} + a\dot{a}h_{ii}\left(\Gamma^j_{jt} - 2\Gamma^i_{it}\right).$$

We now need to compute the three components of the Christoffel symbol Γ^i_{it}, then plug them in and sum over j. We can compute these directly. Using the definition of the Christoffel symbol,

$$\Gamma^i_{ti} = \frac{1}{2}g^{\sigma i}\left(\frac{\partial g_{t\sigma}}{\partial x^i} + \frac{\partial g_{i\sigma}}{\partial x^t} - \frac{\partial g_{ti}}{\partial x^\sigma}\right).$$

Because $g_{\mu\nu}$ is diagonal, $g_{ti} = 0$, so the final derivative vanishes. We also see that g^{ti} vanishes, so we force $\sigma = j$ to only range over spatial coordinates. We then see by again applying diagonal-nes of $g_{\mu\nu}$ that

$$\Gamma^i_{ti} = \frac{1}{2}g^{ji}\left(\frac{\partial g_{tj}}{\partial x^i} + \frac{\partial g_{ij}}{\partial x^t}\right) = \frac{1}{2}g^{ji}\frac{\partial g_{ij}}{\partial x^t} = \frac{1}{2}g^{ii}\frac{\partial g_{ii}}{\partial x^t} = \frac{1}{2}\frac{h^{ii}}{a^2(t)}h_{ii}\frac{\partial}{\partial t}a^2(t) = \frac{1}{2}\frac{2a\dot{a}}{a^2} = \frac{\dot{a}}{a}.$$

Then summing over j yields

$$T_{ii} = \left(\dot{a}^2 + a\ddot{a}\right) h_{ii} + a\dot{a}h_{ii}\left(\frac{3\dot{a}}{a} - \frac{2\dot{a}}{a}\right) = \left(\dot{a}^2 + a\ddot{a}\right) h_{ii} + \dot{a}^2 h_{ii} = \left(2\dot{a}^2 + a\ddot{a}\right) h_{ii}.$$

We therefore have that

$$R_{ii} = T_{ii} + S_{ii} = \left(2\dot{a}^2 + a\ddot{a}\right) h_{ii} + 2h_{ii} = \left(2 + 2\dot{a}^2 + a\ddot{a}\right) h_{ii}.$$

We can then compute the space components of the Ricci scalar:

$$g^{ii}R_{ii} = a^{-2}h^{ii}\left(2 + 2\dot{a}^2 + a\ddot{a}\right) h_{ii} = 3a^{-2}\left(2 + 2\dot{a}^2 + a\ddot{a}\right) = \frac{6}{a^2} + \frac{6\dot{a}^2}{a^2} + \frac{3\ddot{a}}{a}.$$

Finally we compute the time component R_{tt}, which is given by

$$R_{tt} = \partial_\lambda \Gamma^\lambda_{tt} - \partial_t \Gamma^\lambda_{\lambda t} + \Gamma^\lambda_{\lambda\alpha}\Gamma^\alpha_{tt} - \Gamma^\lambda_{t\alpha}\Gamma^\alpha_{\lambda t}.$$

We can simplify this fairly easily. First, since $g_{tt} = -1$, all of its partials vanish, meaning

$$\Gamma^\lambda_{tt} = \frac{1}{2}g^{\sigma\lambda}\left(\frac{\partial g_{t\sigma}}{\partial t} + \frac{\partial g_{t\sigma}}{\partial t} - \frac{\partial g_{tt}}{\partial x^\sigma}\right) = 0.$$

This simplifies it down to

$$R_{tt} = -\partial_t \Gamma^\lambda_{\lambda t} - \Gamma^\lambda_{t\alpha}\Gamma^\alpha_{\lambda t}.$$

We have already computed

$$\Gamma^\lambda_{\lambda t} = \Gamma^t_{tt} + \Gamma^j_{jt} = \Gamma^j_{jt} = \frac{3\dot{a}}{a}.$$

We then have

$$-\frac{\partial}{\partial t}\frac{3\dot{a}}{a} = -3\frac{a\ddot{a} - \dot{a}^2}{a^2}.$$

To compute the other term,

$$\Gamma^\lambda_{t\alpha}\Gamma^\alpha_{\lambda t} = \Gamma^t_{t\alpha}\Gamma^\alpha_{tt} + \Gamma^j_{t\alpha}\Gamma^\alpha_{jt}.$$

Since $\Gamma^t_{t\alpha} = 0$, the $\lambda = t$ component vanishes. We then have

$$\Gamma^\lambda_{t\alpha} = \Gamma^j_{t\alpha} = \Gamma^j_{tt}\Gamma^t_{jt} + \Gamma^j_{tk}\Gamma^k_{jt}.$$

Again $\Gamma^t_{tj} = 0$, so we recover

$$\Gamma^\lambda_{t\alpha}\Gamma^\alpha_{\lambda t} = \Gamma^j_{tk}\Gamma^k_{jt}.$$

To compute this, we will show that $\Gamma^j_{tk} = \Gamma^j_{tj}\delta^j_k$, reducing this to a problem we have already solved. Indeed, using the fact that $g_{\mu\nu}$ is diagonal,

$$\Gamma^j_{tk} = \frac{1}{2}g^{\sigma j}\left(\frac{\partial g_{t\sigma}}{\partial x^k} + \frac{\partial g_{k\sigma}}{\partial t} - \frac{\partial g_{tj}}{\partial x^\sigma}\right) = \frac{1}{2}g^{jj}\left(\frac{\partial g_{tj}}{\partial x^k} + \frac{\partial g_{kj}}{\partial t}\right) = \frac{1}{2}g^{jj}\delta^k_j\frac{\partial g_{jk}}{\partial t} = \Gamma^j_{tj}\delta^j_k.$$

We then have

$$\Gamma^j_{tk}\Gamma^j_{jt} = \Gamma^j_{tj}\Gamma^j_{tj} = \frac{3\dot{a}^2}{a^2}.$$

This yields

$$R_{tt} = 3\left(-\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} - \frac{\dot{a}^2}{a^2}\right) = -\frac{3\ddot{a}}{a}.$$

Therefore, since $g^{tt} = -1$,

$$R = \frac{3\ddot{a}}{a} + \frac{6}{a^2} + \frac{6\dot{a}^2}{a^2} + \frac{3\ddot{a}}{a} = \frac{6}{a}\left(\ddot{a} + \frac{1}{a} + \frac{\dot{a}^2}{a}\right).$$

Problem 2

Problem 2.1. *Verify explicitly that the coordinates*

$$x_1 = (\theta - \theta_p) - \frac{1}{4}(\phi - \phi_p)^2\sin(2\theta_p), \qquad x_2 = (\phi - \phi_p)\sin\theta_p + (\theta - \theta_p)(\phi - \phi_p)\cos\theta_p$$

are locally flat coordinates for the sphere

$$\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2$$

at the point $p = (\theta_p, \phi_p)$.

Solution. We first compute the transformation for the metric on the sphere $g'_{\mu\nu}$ to the new metric $g_{\mu\nu}$ given by

$$g_{\mu\nu} = \frac{\partial x^\alpha}{\partial x^\mu}\frac{\partial x^\beta}{\partial x^\nu}g'_{\alpha\beta}.$$

We don't want to compute θ and ϕ in terms of x_1 and x_2, so we instead compute the inverse:

$$g^{\mu\nu} = (g_{\mu\nu})^{-1} = \left(\frac{\partial x^\alpha}{\partial x^\mu}\frac{\partial x^\beta}{\partial x^\nu}g'_{\alpha\beta}\right)^{-1} = \frac{\partial x^\mu}{\partial x^\alpha}\frac{\partial x^\nu}{\partial x^\beta}g'^{\alpha\beta}.$$

We have to compute all of the partials:

$$\frac{\partial x_1}{\partial\theta} = 1, \qquad \frac{\partial x_1}{\partial\phi} = -\frac{\phi - \phi_p}{2}\sin(2\theta_p),$$

$$\frac{\partial x_2}{\partial\theta} = (\phi - \phi_p)\cos\theta_p, \qquad \frac{\partial x_2}{\partial\phi} = \sin\theta_p + (\theta - \theta_p)\cos\theta_p.$$

Sinnce $g'_{\alpha\beta}$ is diagonal, writing out the summation for $g_{\mu\nu}$ gives

$$g^{\mu\nu} = \frac{\partial x^\mu}{\partial\theta}\frac{\partial x^\nu}{\partial\theta} + \frac{\partial x^\mu}{\partial\phi}\frac{\partial x^\nu}{\partial\phi}\csc^2\theta.$$

We compute each component.

$$g^{11} = \left(\frac{\partial x_1}{\partial \theta}\right)^2 + \left(\frac{\partial x_1}{\partial \phi}\right)^2 \csc^2\theta = 1 + \frac{(\phi-\phi_p)^2}{4}\sin^2(2\theta_p)\csc^2\theta$$

$$g^{12} = g^{21} = \frac{\partial x_1}{\partial \theta}\frac{\partial x_2}{\partial \theta} + \frac{\partial x_1}{\partial \phi}\frac{\partial x_2}{\partial \phi}\csc^2\theta$$

$$= (\phi-\phi_p)\cos\theta_p + \left(-\frac{\phi-\phi_p}{2}\sin(2\theta_p)\right)(\sin\theta_p + (\theta-\theta_p)\cos\theta_p)\csc^2\theta$$

$$g^{22} = \left(\frac{\partial x_2}{\partial \theta}\right)^2 + \left(\frac{\partial x_2}{\partial \phi}\right)^2 \csc^2\theta$$

$$= (\phi-\phi_p)^2\cos^2\theta_p + \left(\sin^2\theta_p + (\theta-\theta_p)^2\cos^2\theta_p + 2(\theta-\theta_p)\sin\theta_p\cos\theta_p\right)\csc^2\theta.$$

Now if we evaluate $g^{\mu\nu}$ at p, we obtain

$$g^{11} = 1, \qquad g^{21} = g^{12} = 0, \qquad g^{22} = \sin^2\theta_p\csc^2\theta_p = 1.$$

Therefore, $g^{\mu\nu}(p) = \delta^\nu_\mu$. Since the inverse to the identity matrix is the identity matrix, $g_{\mu\nu} = \delta^\nu_\mu$.

Now we show the first derivatives vanish. We will show it suffices to show that the derivatives of $g^{\mu\nu}$ with respect to the old coordinates vanish at p. This suffices because, writing y^μ for the coordinates θ, ϕ and x^μ for the new coordinates x_1, x_2, the chain rule yields

$$\frac{\partial g^{\mu\nu}}{\partial y^\alpha} = \frac{\partial g^{\mu\nu}}{\partial x^\lambda}\frac{\partial x^\lambda}{\partial y^\alpha} \implies \frac{\partial g^{\mu\nu}}{\partial x^\lambda} = \frac{\partial g^{\mu\nu}}{\partial y^\alpha}\frac{\partial y^\alpha}{\partial x^\lambda}.$$

So if the derivatives with respect to the old coordinates vanish at p, then so will the ones for the new metric. This implies that those for $g_{\mu\nu}$ vanish as well because

$$g^{\mu\rho}g_{\rho\nu} = \delta^\nu_\mu \implies \frac{\partial}{\partial x^\lambda}(g^{\mu\rho}g_{\rho\nu}) = \frac{\partial}{\partial x^\lambda}\delta^\nu_\mu = 0.$$

Expanding the product rule yields

$$\frac{\partial g^{\mu\rho}}{\partial x^\lambda}g_{\rho\nu} + \frac{\partial g_{\rho\nu}}{\partial x^\lambda}g^{\mu\rho} = 0.$$

If $\frac{\partial g^{\mu\rho}}{\partial x^\lambda}$ vanishes at p, then we have

$$\frac{\partial g_{\rho\nu}}{\partial x^\lambda}g^{\mu\rho} = 0.$$

Since the metric is invertible,

$$\frac{\partial g_{\rho\nu}}{\partial x^\lambda} = 0.$$

We now conclude by computing more partials and showing they all vanish at p. First the θ-derivatives:

$$\frac{\partial g^{11}}{\partial \theta} = \frac{(\phi-\phi_p)^2}{4}\sin^2(2\theta_p)\left(-2\csc^2\theta\cot\theta\right)$$

$$\frac{\partial g^{12}}{\partial \theta} = \left(-\frac{\phi-\phi_p}{2}\sin(2\theta_p)\right)\left(-2\sin\theta_p\csc^2\theta\cot\theta + \cos\theta_p\left(\csc^2\theta - 2(\theta-\theta_p)\csc^2\theta\cot\theta\right)\right)$$

$$\frac{\partial g^{22}}{\partial \theta} = -2\sin^2\theta_p\csc^2\theta\cot\theta + 2(\theta-\theta_p)\cos^2\theta_p\csc^2\theta - 2(\theta-\theta_p)^2\cos^2\theta_p\csc^2\theta\cot\theta$$

$$+2\sin\theta_p\cos\theta_p\left(\csc^2\theta - 2(\theta-\theta_p)\csc^2\theta\cot\theta\right).$$

The first two vanish at p due to the $\phi - \phi_p$ in the prefactor. The final one vanishes because if we cancel all the terms with $\theta - \theta_p$, then we are left with

$$-2\sin^2\theta_p\csc^2\theta_p\cot\theta_p + 2\sin\theta_p\cos\theta_p\csc^2\theta_p = 2\left(\frac{\sin\theta_p\cos\theta_p}{\sin^2\theta_p} - \frac{\sin^2\theta_p\cos\theta_p}{\sin^3\theta_p}\right) = 0.$$

Finally the ϕ-derivatives:

$$\frac{\partial g^{11}}{\partial\phi} = \frac{\phi-\phi_p}{2}\sin^2(2\theta_p)\csc^2\theta$$

$$\frac{\partial g^{12}}{\partial\phi} = \cos\theta_p - \frac{\sin(2\theta_p)}{2}\left(\sin\theta_p + (\theta-\theta_p)\cos\theta_p\right)\csc^2\theta$$

$$\frac{\partial g^{22}}{\partial\phi} = 2(\phi-\phi_p)\cos^2\theta_p.$$

The first and final ones vanish at p due the the $\phi - \phi_p$ prefactor, and the second one vanishes at p because

$$\cos\theta_p - \frac{\sin(2\theta_p)}{2}\left(\sin\theta_p + (\theta_p-\theta_p)\cos\theta_p\right)\csc^2\theta_p = \cos\theta_p - \frac{\sin(2\theta_p)}{2}\sin\theta_p\csc^2\theta_p$$

$$= \cos\theta_p - \frac{\sin(2\theta_p)}{2\sin\theta_p}.$$

Double angle identities yield

$$= \cos\theta_p - \frac{2\sin\theta_p\cos\theta_p}{2\sin\theta_p} = \cos\theta_p - \cos\theta_p = 0.$$